Kinetic Sight CosmOS: A Multimodal Web Interface for Ergonomic Spatial Exploration

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Abstract

Touchless interfaces offer highly immersive experiences for educational and public contexts, but they frequently suffer from critical usability issues. Pure gaze tracking often leads to unintentional selections, while prolonged mid-air hand gestures cause severe physical fatigue. Furthermore, existing state-of-the-art spatial solutions often rely either on traditional WIMP mechanics or expensive, specialized hardware with infrared sensors. To address these limitations, we present Kinetic Sight CosmOS: a web-based multimodal interface for exploring an interactive 3D Solar System. Our system explicitly decouples interactions across three complementary modalities: Gaze Tracking for passive targeting, Hand Gestures for kinetic activation, and Speech Recognition for parameter manipulation and system control. This architecture is supported by a novel Distributed Sensor Setup using decentralized, everyday 2D RGB cameras. To evaluate the system, we conducted a within-subjects user study across three incremental tasks using a mixed-methods approach. We implemented a custom background data logger to capture objective performance metrics and paired it with structured questionnaires for subjective workload assessment. Results demonstrated rapid user adaptation, a high System Usability Scale score, and a near-total elimination of false-positive activations. Furthermore, workload evaluations revealed minimal physical demand, confirming the ergonomic viability of the design. Finally, data triangulation between the objective logs and subjective scores demonstrated the system's high resilience to browser-based Speech-to-Text limitations; users maintained very low frustration levels even when experiencing voice recognition errors, validating the robustness of the kinetic approach for future educational and entertainment applications.

CCS Concepts

• **Human-centered computing → Interaction techniques**; *Virtual reality*.

Keywords

Multimodal Interaction, Human-Computer Interaction, Touchless Interface, Gaze Tracking, Gestures, Usability

1 Introduction

The evolution of Human-Computer Interaction has increasingly moved towards touchless, spatial paradigms, particularly in fields like Virtual Reality (VR) and Edutainment. These interfaces offer highly immersive, intuitive, and hygienic experiences, making them ideal for public exhibitions and interactive learning environments. However, relying solely on a single input modality often introduces severe usability bottlenecks that hinder prolonged adoption.

Pure gaze-based interfaces typically suffer from the widely documented "Midas Touch" problem [10]. Because human vision requires constantly scanning the environment, using eye position as a direct substitute for a traditional cursor is unworkable; users cannot look at an item without unintentionally issuing a command. In his foundational research, Jacob observed the continuous physiological jitter and saccades of the human eye, noting that systems had to rely on artificial "dwell time" thresholds or physical buttons to separate passive viewing from intentional selection [10]. Conversely, interfaces relying entirely on mid-air hand tracking frequently induce the "Gorilla Arm" effect, a condition characterized by rapid muscular fatigue and a feeling of heaviness in the upper limbs [7]. To scientifically quantify this limitation, Hincapié-Ramos et al. developed "Consumed Endurance", a biomechanical metric evaluated via skeleton-tracking cameras, which definitively demonstrated the high physical exertion and rapid fatigue associated with continuous, unsupported arm movements [7].

While state-of-the-art commercial headsets successfully mitigate these issues by fusing gaze and micro-gestures, they rely on expensive, proprietary infrared hardware. Conversely, accessible web-based astronomical tools remain strictly anchored to traditional WIMP (Windows, Icons, Menus, Pointer) mechanics, sacrificing spatial immersion.

To address these limitations, we propose **Kinetic Sight CosmOS**, an experimental multimodal web interface that synergizes Gaze, Kinetics (Hand Gestures), and Voice Recognition. By explicitly assigning passive targeting to the eyes, kinetic activation to the hands, and complex parameter manipulation to the voice, we distribute the cognitive and physical load across multiple sensory channels. Furthermore, we achieve this through a novel "Distributed Sensor Setup," leveraging decentralized, everyday 2D RGB cameras to democratize spatial interaction.

2 Related Work

To define our interaction model, we analyzed existing literature and commercial applications, identifying two major contrasting paradigms that our project aims to bridge and improve upon.

2.1 Spatial Computing and the Dwell-Time Dilemma

Recent progress in spatial computing, notably Apple's Vision Pro headset, has popularized multimodal interaction paradigms that fuse vision and micro-gestures. According to Apple's official design guidelines [2], the system's primary interaction relies on users targeting a virtual object with their eyes and activating it through an "indirect gesture", such as a tap, performed while their hands rest comfortably. From an HCI perspective, this approach elegantly solves a classic issue in pure eye-tracking: the reliance on "dwell-time" mechanics (staring at an object for a set duration to select it), which often leads to the Midas Touch problem and false positives. Academic research strongly supports this multimodal fusion; for instance, a recent study by Wang et al. demonstrated that combining gaze, gesture, and speech in Augmented Reality (AR) significantly improves interaction efficiency over bimodal or single-input systems [20]. However, both commercial headsets and these experimental AR systems rely on expensive, specialized hardware with infrared tracking and head-mounted displays. Our project democratizes this multimodal paradigm by implementing it directly in a standard web browser, using decentralized, everyday 2D RGB cameras and local machine learning inference to strictly separate passive targeting from kinetic activation.

2.2 Web-Based Edutainment and Traditional Interfaces

In the context of astronomical edutainment, traditional desktop applications have proven highly effective for education. For example, studies on desktop software like Stellarium show that interactive multimedia significantly enhances students' understanding and motivation when learning about the solar system [1]. Similarly, NASA's "Eyes on the Solar System" represents a state-of-the-art interactive 3D web tool built on real mission data [14]. While visually impressive and educationally valuable, the interaction models of these tools remain strictly anchored to traditional WIMP (Windows, Icons, Menus, Pointer) mechanics, requiring heavy reliance on mouse clicks and dragging. Kinetic Sight CosmOS overcomes this physical barrier by proposing a completely touchless environment. By introducing a multimodal approach, users navigate the same rich astronomical data physically and vocally, greatly reducing the physical fatigue associated with traditional desktop hardware or gesture-only spatial interfaces.

3 System Overview and Interaction Model

To provide a clear understanding of the user experience, this section details the application interface, the initial setup phase, and the comprehensive dictionary of multimodal commands available to the user.

3.1 Interface and Gaze Calibration Phase

Upon launching Kinetic Sight CosmOS, the user is presented with a 3D rendering of the solar system. The interface is designed to be minimal, featuring a transparent Heads-Up Display (HUD) that provides real-time feedback on the status of the three tracking modalities (Gaze, Hand, Voice) and the currently targeted celestial body.

Overcoming Webcam Gaze Limitations. Before entering the 3D environment, the system prompts the user with an optional, but strongly recommended, gaze calibration phase. Scientific literature widely documents that unconstrained eye-tracking via standard RGB webcams is inherently less accurate than specialized infrared equipment [18]. This poor accuracy is primarily due to diverse local environments, variable ambient lighting, and differing human facial features. Furthermore, unlike commercial eye trackers that build precise 3D models of the eye, webcam-based solutions lack explicit 3D reasoning. Instead, they must infer complex mappings between 2D pixel features and screen coordinates, a process heavily disrupted by any changes in the 3D position and rotation of the user's head relative to the camera [18].

To mitigate these severe hardware limitations and establish a reliable mathematical mapping between facial features and screen coordinates, users are highly encouraged to complete a 13-point calibration process. By clicking on specific on-screen targets while fixing their gaze upon them, the system trains a local regression model to correlate pupil positions with display coordinates, ensuring the robust passive targeting required for our multimodal interface.

3.2 The Multimodal Dictionary

Once calibrated, the user navigates the application entirely hands-free. The interaction model strictly separates the targeting phase from the execution phase to prevent unintentional selections.

1. Passive Targeting (Gaze): Users simply look at a planet. The system visually highlights the celestial body being fixated upon, confirming it as the active target without triggering any immediate action.

2. Kinetic Activation and Navigation (Hand Gestures): Manual gestures serve as the primary kinetic triggers:

- **Pinch:** Touching the thumb and index finger together "grabs" the currently gazed-at planet, unlocking contextual voice commands.
- **Vertical Scroll:** Moving the hand up or down dynamically scrolls through the text inside the planet's detail panel.
- **Lateral Swipe:** A swift horizontal hand strike acts as a "Back" or "Exit" command to close the detail panel and return to the solar system overview.

3. Property Manipulation and Control (Voice): Speech recognition is categorized into two distinct levels to maximize reliability:

- **Global Commands (Always Active):** Users can alter the simulation time ("Faster", "Slower", "Stop", "Resume"), jump directly to specific planets ("Show Jupiter", "Open Earth"), or navigate interfaces ("Back").
- **Contextual Commands (Active only while Pinching):** To prevent false positives while speaking naturally, destructive or modifying commands are only accepted while the user is actively "holding" a planet via the pinch gesture. These include scale modifications ("Enlarge", "Shrink", "Reset"), visual property changes (e.g., "Red", "Blue", "Gold", "White"), and requests for astrophysical data ("Details", "Information").

4 System Design and Implementation

The architecture of Kinetic Sight CosmOS follows a modular design, built entirely on a web-native stack to ensure cross-platform accessibility and low-latency multimodal interaction without requiring external plugins.

4.1 Technical Stack and Asset Sourcing

The system is implemented using **HTML5**, **CSS3**, and **JavaScript (ES6+)**, leveraging the local GPU via WebGL for 3D rendering. To ensure high visual fidelity, all high-resolution planetary textures and atmospheric maps were sourced from the Solar System Scope website [9]. To guarantee a zero-latency educational experience and high system responsiveness, all astrophysical parameters and textual descriptions, strictly sourced from the **NASA Planetary Fact Sheets** [13] and the **NASA Solar System Exploration** portal [15] to ensure scientific accuracy, are hardcoded into a comprehensive `planetsConfig` object array. This architecture avoids external API queries, significantly reducing network overhead and ensuring standalone reliability. Additionally, the vocal interaction engine leverages the `webkitSpeechRecognition` interface, meaning the application is currently optimized specifically for Chromium-based browsers (e.g., Google Chrome) to ensure stable, browser-native speech processing. Finally, data analysis was conducted using a dedicated **Python** pipeline (utilizing Pandas and Matplotlib) to process the experimental logs generated during the user study. The complete source code, including the interaction engine and the analysis scripts, is publicly available on GitHub [11].

4.2 3D Engine and Rendering

The planetary environment is rendered using **Three.js**. Each celestial body is modeled as a high-resolution sphere utilizing physically based rendering (PBR) materials. Instead of computationally heavy custom shaders, atmospheric and lighting effects (such as the Sun's coronal glow and planetary clouds) are efficiently achieved through advanced texture mapping and additive blending, easily maintaining a strict high-frequency rendering loop (60 FPS). For spatial targeting, a **Raycaster** utility translates the 2D gaze coordinates provided by the eye tracker into 3D world space. To optimize performance, the raycaster evaluates the geometric distance from the ray to the center of each celestial body rather than performing complex mesh intersections, rapidly identifying the user's focal target.

4.3 Multimodal Fusion Logic

While the previous section outlined the user-facing interaction model, the technical innovation lies in the data fusion engine (implemented within `app.js`). Following foundational multimodal research by Oviatt [17], our architecture primarily leverages **complementarity** for spatial navigation: passive targeting (gaze) and active execution (kinetics) process synergistic chunks of information that are merged seamlessly. However, to address the inherent inaccuracies of webcam-based eye tracking, the system also implements **equivalence** by allowing users to target planets directly via global voice commands. Providing this flexible multimodal alternative allows users to intuitively bypass an error-prone input mode, significantly increasing the overall robustness of the interface [17]. To enable this synergistic interaction, the engine acts as a real-time synchronization hub, employing a state-machine logic to fuse three concurrent input streams:

- **Targeting Stream (Eye Tracking):** This process is handled by `WebGazer.js`. To mitigate natural eye jitter and tracking noise, the raw 2D screen coordinates are processed through a linear interpolation filter (low-pass). These smoothed coordinates are then normalized and fed into the Three.js raycaster to continuously update the `hoveredGroup` state variable without jitter.
- **Action Stream (Hand Tracking):** Handled by `MediaPipe Hands`. The engine evaluates the 3D landmark array at every frame. A "Pinch" boolean state is toggled by calculating the Euclidean distance between the thumb (landmark 4) and index finger (landmark 8) tips against a strict threshold. For dynamic gestures, the engine buffers the wrist's coordinate history (landmark 0), evaluating spatial deltas to trigger scroll or swipe events. While the architecture natively supports multiplexing both gaze and hand tracking through a single webcam stream, to maximize tracking reliability and physically eliminate arm fatigue, we introduce an optimal **Distributed Sensor Setup**. This setup processes skeletal data via a secondary camera stream (e.g., a smartphone), completely decoupling the hand-tracking coordinate space from the facial gaze-tracking space.
- **Modality Stream (Voice):** Managed by the native Web Speech API using `VoiceManager.js`. The engine utilizes the kinetic input to gate the audio processing: continuous speech recognition runs asynchronously, but the fusion hub uses the current kinetic state (`isPinchingGlobally`) as a conditional boolean gate. If the gate is open (a planet is actively grabbed), the parser routes the final audio transcripts to the modifier functions of the targeted 3D object. If closed, only overarching system commands are parsed. This strict logical dependency between hand state and voice execution algorithmically prevents false-positive activations.

4.4 Multisensory Feedback and System Status

To minimize cognitive load and provide clear system state awareness, Kinetic Sight CosmOS implements a real-time multisensory feedback loop, specifically designed to bridge Norman's **Gulf of Evaluation** [16]. By providing immediate acoustic and visual confirmation, the system minimizes the cognitive effort required by the user to interpret the current state of the interface, ensuring clear feedback for every interaction across all modalities:

- **Auditory Cues (`AudioManager`):** The system utilizes the Web Audio API to generate procedural waveforms that provide non-visual confirmation of kinetic states. Distinct tones are triggered for specific events: a high-pitch sine wave (800Hz) for target hovering, a low-pitch square wave (200Hz) to signal a successful "grab," and a descending sawtooth wave (150Hz) upon release. Additionally, a Text-to-Speech (TTS) engine provides brief contextual announcements (e.g., confirming opening planet details), ensuring users are constantly aware of the system's background processes.
- **Visual State Indication and Kinetic Lock:** The interaction engine employs linear interpolation (LERP) with a 0.1 factor to ensure all visual 3D transitions are fluid. When a celestial body is aimed at, it scales up by 20% and activates its emissive material. Upon a successful pinch, the system provides immediate physics-based feedback: the planet's orbital movement is deliberately halted, it scales to 160% of its original size, and it changes its emissive glow to a distinct red. This combination of pausing the orbital animation and illuminating the mesh provides the user with an unmistakable sensation of physically holding the object.

- **Voice Status Monitoring:** To reassure the user that the system is ready to accept commands, the UI provides constant visibility of the microphone status. The "Voice" indicator remains consistently green to signal that the Web Speech API is continuously listening. Furthermore, successful voice commands that modify planetary properties (e.g., color changes) are immediately confirmed by a dedicated procedural audio tone (500Hz triangle wave), seamlessly closing the interaction loop.

4.5 Data Logging Architecture

To enable rigorous quantitative analysis, we implemented a non-intrusive `DataLogger` integrated directly into the main interaction loop within `app.js`. This component serves as the bridge between system performance and experimental validation.

The logger is managed via specific researcher-controlled voice commands (e.g., *"start task"* and *"finish task"*), which toggle the `testActive` state. While a task is active, the system monitors the interaction state at 60 FPS, recording the following metrics:

- **Temporal Efficiency:** Task duration is calculated by capturing high-resolution timestamps at the initiation and conclusion of each task.
- **Kinetic Accuracy:** The system tracks *Missed Pinches*, defined as activation attempts where a pinch gesture is detected while the gaze-raycaster does not intersect with any celestial body.
- **Targeting Precision:** The logger counts *Total Grabs*, reflecting successful synchronizations between gaze-targeting and hand-activation.
- **Speech Reliability:** The engine monitors *Voice Fails*, incrementing an error counter whenever a speech transcript is finalized but fails to match the expected command dictionary.

Upon completion, the accumulated data is formatted into a log entry, stored in the browser's `localStorage`, and exported via a generated `.txt` file (using a `Blob` and an automated download trigger) for batch processing through our Python-based analysis pipeline.

5 Methodology and User Study

To rigorously validate the proposed multimodal paradigm, we conducted a usability study involving 19 participants (Age: < 18 through 45+, with varying degrees of VR/Voice interface experience). The primary goal of this usability test is to evaluate the Learnability, Efficiency, and Ergonomics of the Kinetic Sight CosmOS interface. Instead of comparing the system to traditional peripherals (like a mouse), the study employs a Within-Subjects Design to measure how quickly users adapt to a completely touchless, multimodal paradigm.

5.1 Experimental Protocol and Task Design

The tasks are designed incrementally to assess the system's ability to prevent common HCI issues, such as unintentional activations (false positives) and physical fatigue. Participants were asked to complete the following three sequential tasks:

- **Task 1: Basic Exploration.** The user must locate Jupiter using their gaze and perform a "Pinch" gesture to grab it. While holding the pinch, they must say "Details" to open the information panel, and finally say "Back" (or perform a lateral "Swipe") to close it. This task evaluates fundamental hand-eye-voice coordination. It proves our solution to the "Midas Touch" problem: targeting (eyes), holding (hand), and opening (voice) are completely decoupled, making unintentional activations virtually impossible.
- **Task 2: Multimodal Synergy.** The user must find Mars, grab it with a Pinch, and hold it. While holding the planet, they must use voice commands, saying "Enlarge" to increase its scale, and "Blue" to change its color. Finally, they release the pinch to exit. This task measures the reduction of Cognitive Load. By delegating complex property modifications (size, color) to the vocal channel while the hand is simply holding the object, the system avoids forcing the user to learn complex or unnatural hand gestures.
- **Task 3: Fine Control & Data Navigation.** The user grabs Earth and says "Details" to open its data panel. By simply moving their hand vertically, they scroll down the text, locate a specific piece of data (e.g., "Escape Velocity"), read it aloud, and finally say "Back" to close the panel. This task is crucial for evaluating Physical Ergonomics. By allowing users to scroll with small, relaxed wrist movements and exit using a simple vocal command, this task gathers data to evaluate physical exertion, avoiding the fatigue typical of traditional mid-air interfaces.

5.2 Quantitative Data Collection

Throughout all three tasks, the custom background Data Logger automatically recorded the quantitative performance metrics: Time on Task, Missed Pinches, Total Grabs, and Voice Fails. By having users perform these increasingly complex tasks sequentially within a single session, the collected data empirically demonstrated the system's rapid intra-session learning curve, multimodal reliability, and high usability.

Immediately following the successful completion of the practical tasks and the automatic logging of their quantitative performance, participants were directed to the qualitative evaluation phase to complete structured subjective questionnaires.

5.3 Qualitative Evaluation Instruments

Immediately following the practical session, participants completed a multi-section structured questionnaire designed to capture subjective insights and psychological workload [12]. The decision to administer the survey post-test was strategic: it ensured that responses were grounded in the immediate cognitive and physical memory of the interaction, particularly regarding fatigue and frustration levels.

The evaluation was divided into four key dimensions:

(1) **Demographics and Prior Experience:** This section contextualized the results by mapping participants' age groups and their familiarity with VR, gesture-based interfaces, and voice assistants. This allowed us to evaluate the system's learnability across different expertise levels.

(2) **System Usability Scale (SUS):** To obtain a reliable, global view of subjective usability, we utilized the industry-standard SUS [5]. Comprised of a simple, ten-item Likert scale, this tool provides a "quick and dirty" yet highly robust composite measure of the overall usability of the system. It was selected because it yields a single global score (0 to 100) and is explicitly designed for low-cost, reliable assessments across different contexts.
(3) **NASA Task Load Index (NASA-TLX):** Given our focus on mitigating physical fatigue and cognitive overload, we administered an adapted subset of the NASA-TLX [6]. We focused specifically on the three sub-scales most critical for touchless spatial interfaces: *Mental Demand*, *Physical Demand*, and *Frustration*. In our study, this was the primary tool for quantifying the ergonomic success of the distributed sensor setup and identifying specific bottlenecks in the multimodal fusion.
(4) **Future Scenarios and Barriers:** The final section used a mixed-method approach (multiple-choice and open-ended questions) to identify potential use cases (e.g., museums, schools) and the primary adoption barriers, such as environmental lighting or social discomfort with voice commands.

6 Results and Data Analysis

6.1 Learnability and Efficiency

Objective logs collected from 19 participant sessions demonstrated a rapid adaptation to the multimodal paradigm. As illustrated in Fig. 1, the average time required to complete the tasks decreased consistently. While Task 1 (Basic Exploration) and Task 2 (Multimodal Synergy) required an average of 36.2s and 35.1s respectively, completion time dropped significantly to 25.6s in Task 3. This indicates high system learnability: once users internalized the core mechanics in the initial stages, executing complex data navigation became highly efficient.

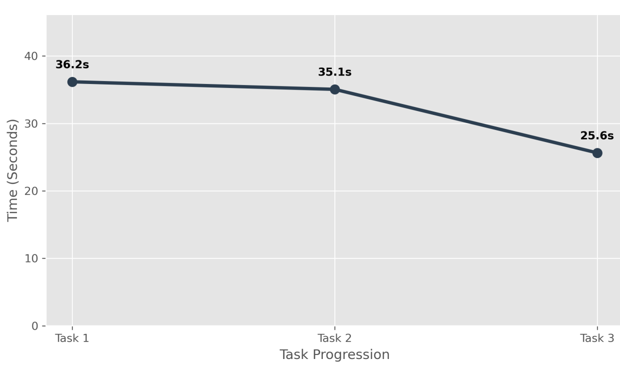

Figure 1: Learning Curve: Average Time on Task

6.2 Error Prevention (Midas Touch Resolution)

A primary goal of Kinetic Sight CosmOS was the elimination of unintentional activations. Fig. 2 reveals a sharp decline in "Missed Pinches" (kinetic activations attempted without a valid gaze target), dropping from an average of nearly 2.8 per user in Task 1 to roughly 0.7 by Task 3. Users quickly adapted to the sequence of aiming with their eyes before engaging their hands. The qualitative survey corroborated this trend, with participants frequently highlighting the system's robustness against accidental selections. Users consistently reported feeling comfortable observing the virtual environment without the fear of triggering unintended actions. From an HCI perspective, this collective feedback empirically demonstrates the successful mitigation of the Midas Touch problem through the explicit decoupling of the targeting and activation modalities.

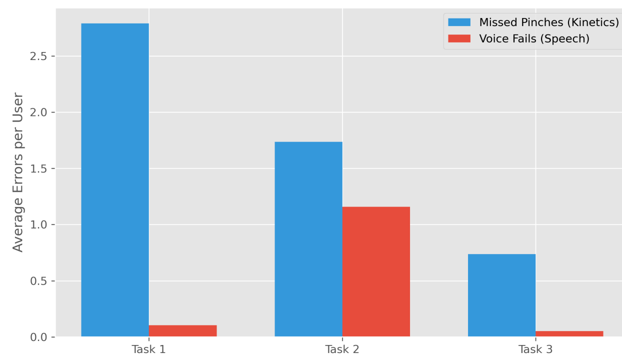

Figure 2: Multimodal Error Reduction over the three tasks

6.3 Usability and Workload Assessment (SUS and NASA-TLX)

To evaluate the overall user experience and system viability, we deeply analyzed the subjective responses from the System Usability Scale (SUS) and the NASA Task Load Index (NASA-TLX). All quantitative logs and qualitative survey responses were exported as raw datasets and processed through a custom Python analysis pipeline utilizing the Pandas library. This automated approach ensured that raw Likert-scale responses were algorithmically normalized and aggregated according to their respective official scoring formulas, preventing manual calculation errors.

Global Usability (SUS): The system achieved an average SUS score of 77.4. All qualitative survey responses were processed through a custom Python analysis pipeline [11] utilizing the Pandas library to implement the standard SUS scoring algorithm:

$$SUS = \left(\sum_{n=odd} (R_n - 1) + \sum_{n=even} (5 - R_n) \right) \times 2.5 \qquad (1)$$

where R_n represents the user's response to the n^{th} question. According to the empirical evaluation by Bangor et al. [4], a score of 68 represents the average baseline for usability. Furthermore, utilizing the validated adjective rating scale [3], our score of 77.4 positions Kinetic Sight CosmOS in the "Good" category. This is a highly significant outcome for an experimental, entirely touchless interface. It empirically demonstrates that despite the lack of tactile feedback, the absence of traditional WIMP mechanics, and the reliance on standard consumer webcams, the multimodal architecture remains highly approachable, consistent, and functionally coherent for everyday users.

Psychological and Physical Workload (NASA-TLX): The multi-dimensional NASA-TLX results (Fig. 3) provided critical scientific validation for our ergonomic and cognitive hypotheses:

- **Physical Demand (2.2/7):** This exceptionally low score is the most crucial finding of the workload assessment. It definitively proves that the Distributed Sensor Setup, which allows users to rest their elbows comfortably on a desk while performing micro-gestures (Pinch/Vertical Scroll), successfully eradicates the severe muscular fatigue and physical strain traditionally associated with mid-air spatial interfaces.
- **Mental Demand (2.8/7):** While slightly higher than the physical exertion, a score of 2.8 remains objectively low. It indicates that explicitly distributing tasks across three distinct sensory channels (aiming with eyes, grabbing with hands, editing with voice) effectively balances the cognitive load. Rather than overwhelming the user, this separation of concerns makes complex 3D navigation mentally manageable. As extensively documented in HCI literature, continuous speech production heavily taxes verbal working memory [19]. By restricting voice inputs to short, contextual commands (e.g., “Red”, “Details”) while the hands manage spatial holding, we successfully prevent cognitive overload.
- **Frustration (2.3/7):** The minimal frustration reported confirms the success of the system’s strict state-machine logic. By ensuring that vocal modifiers only trigger during an active kinetic hold, the architecture successfully prevented the cascading false-positive errors that typically plague unimodal gaze or voice systems.

Synthesizing the Findings: Drawing conclusions from these combined metrics, the data unequivocally proves that multimodal decoupling is a highly effective solution to the fundamental bottlenecks of touchless HCI. By systematically assigning passive targeting to gaze, active triggers to manual kinetics, and complex parameter manipulation to speech, Kinetic Sight CosmOS achieves a rare balance: it is physically relaxing to use (validated by NASA-TLX), highly learnable for novices (validated by SUS and the Time-on-Task learning curve), and resilient against unintentional user errors (validated by objective logger data).

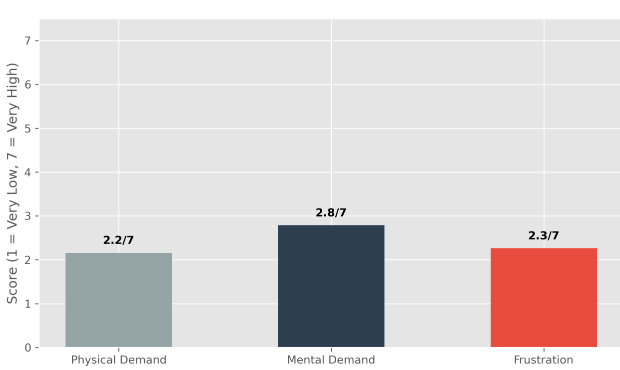

Figure 3: NASA-TLX Workload Assessment

7 Discussion and Future Scenarios

7.1 Data Triangulation: System Resilience to Errors

To identify potential usability bottlenecks, we triangulated the objective "Voice Fails" from the Data Logger with the subjective "Frustration" scores from the NASA-TLX (Fig. 4). Contrary to typical HCI assumptions, the scatter plot and trendline reveal a negligible, slightly negative correlation. Remarkably, users who experienced the highest number of voice recognition errors (e.g., 4 or 5 fails) still reported very low frustration levels (2/7). This suggests that the "pinch-to-command" interaction is so physically effortless that needing to repeat a voice command due to browser-based Speech-to-Text limitations does not significantly impact the user’s psychological workload.

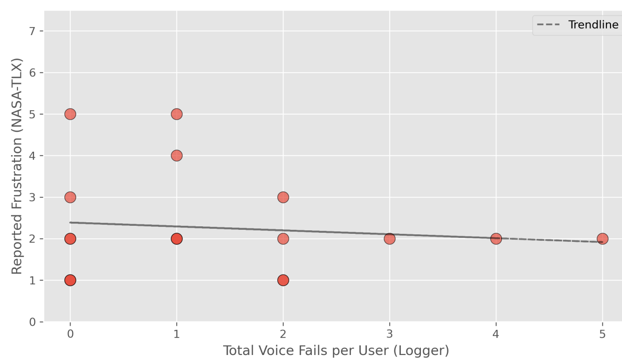

Figure 4: Triangulation analysis

7.2 User Reception and Adoption Barriers

Participants predominantly selected "Home Entertainment or Gaming" (9 votes) and "Educational tools" (8 votes) as the most suitable environments for Kinetic Sight CosmOS (Fig. 5).

Also, qualitative feedback was overwhelmingly enthusiastic, with users likening the experience to having "superpowers" or using "a real-life Sci-Fi interface."

However, the survey also identified practical adoption barriers. The primary limitation cited was the complex physical setup, specifically the need for optimal lighting and dual cameras. Additionally, some users noted the interface initially "requires too much concentration compared to a simple mouse", highlighting the need for enhanced visual onboarding in future iterations.

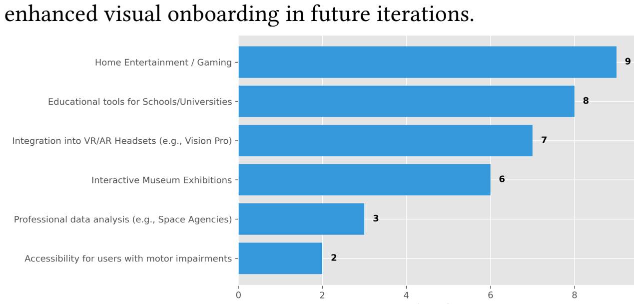

Figure 5: User preferences for the future adoption

7.3 Future Work

While Kinetic Sight CosmOS successfully mitigates physical fatigue and the Midas Touch problem, several avenues remain for enhancing the multimodal experience. Currently, the system relies on a strict user-initiative dialogue strategy. To better address the browser-based Speech-to-Text limitations identified during our evaluation, future iterations could explore Mixed-Initiative Interaction paradigms [8]. By enabling the system to proactively request clarification upon detecting low-confidence speech transcripts, thereby implementing an active error-repair loop, it would be possible to further bridge the Gulf of Evaluation [16] and create a more conversational, cooperative dynamic.

Furthermore, future research could integrate Socio-Affective Computing capabilities to monitor user engagement in real-time. By leveraging the existing webcam stream to extract Facial Action Units (FACS), such as the Brow Lowerer associated with confusion or frustration, the system could dynamically adjust its parameters (e.g., slowing down planetary rotation or providing visual hints). This transition from a purely reactive interface to an emotionally-aware, adaptive system would maximize the educational impact and ensure prolonged usability across diverse user demographics.

8 Conclusion

In this report, we presented **Kinetic Sight CosmOS**, a novel web-based multimodal interface designed to democratize touchless spatial interaction for astronomical exploration. By explicitly decoupling interaction across three sensory channels, assigning passive targeting to gaze, kinetic triggers to hand gestures, and complex parameter manipulation to speech, we successfully addressed long-standing HCI bottlenecks such as the Midas Touch problem and physical fatigue.

Our experimental results, derived from a within-subjects study with 19 participants, provide strong empirical evidence for the system's efficacy. The objective data demonstrated a steep learning curve and a near-total elimination of unintentional activations. Simultaneously, the subjective evaluations via SUS (77.4) and NASA-TLX confirmed that the system is highly usable, intuitive, and physically relaxing. Notably, the implementation of a **Distributed Sensor Setup** proved to be a key ergonomic innovation, allowing for robust tracking using consumer-grade hardware while maintaining a low physical demand (2.2/7).

While limitations in browser-based speech recognition remain, our triangulation analysis suggests that the overall interface design is resilient enough to maintain high user satisfaction even during technical failures. Ultimately, Kinetic Sight CosmOS proves that sophisticated, immersive, and hygienic touchless experiences can be delivered through standard web technologies, opening new pathways for accessible edutainment in museums, schools, and home environments.

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